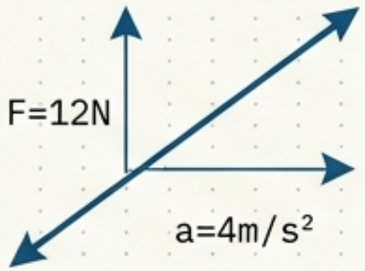
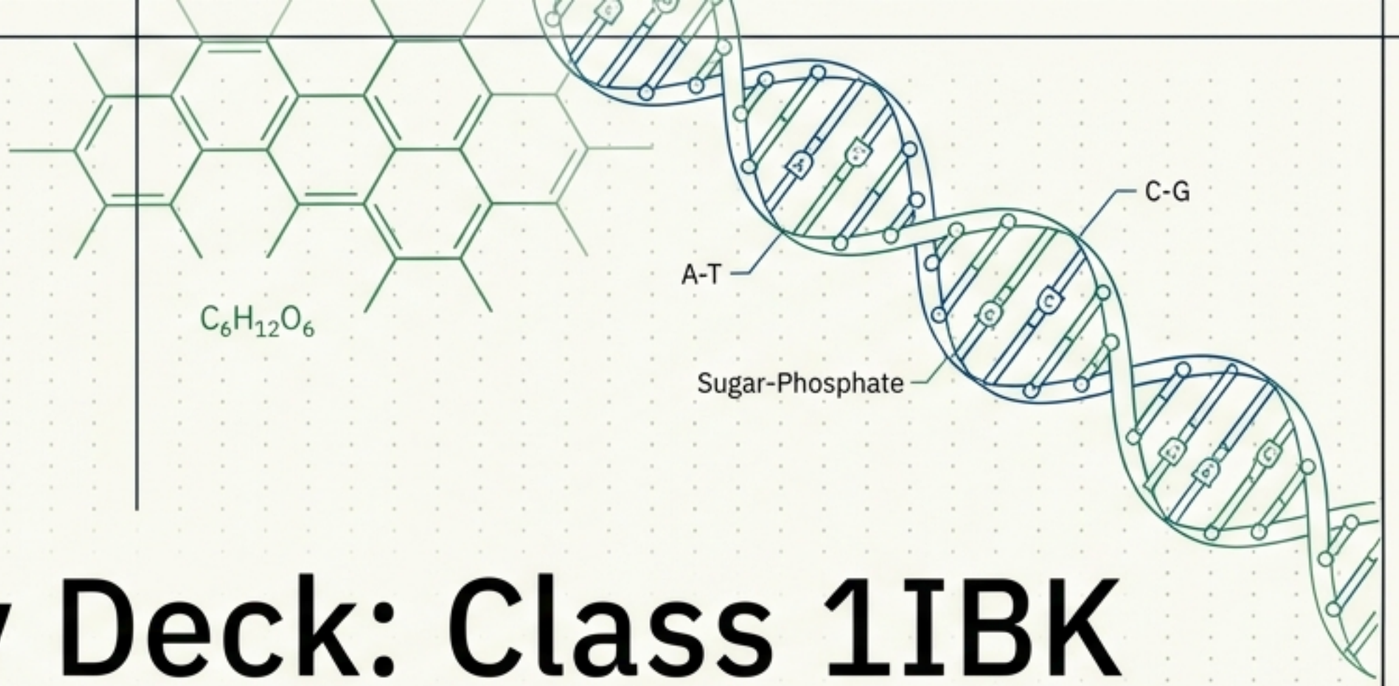
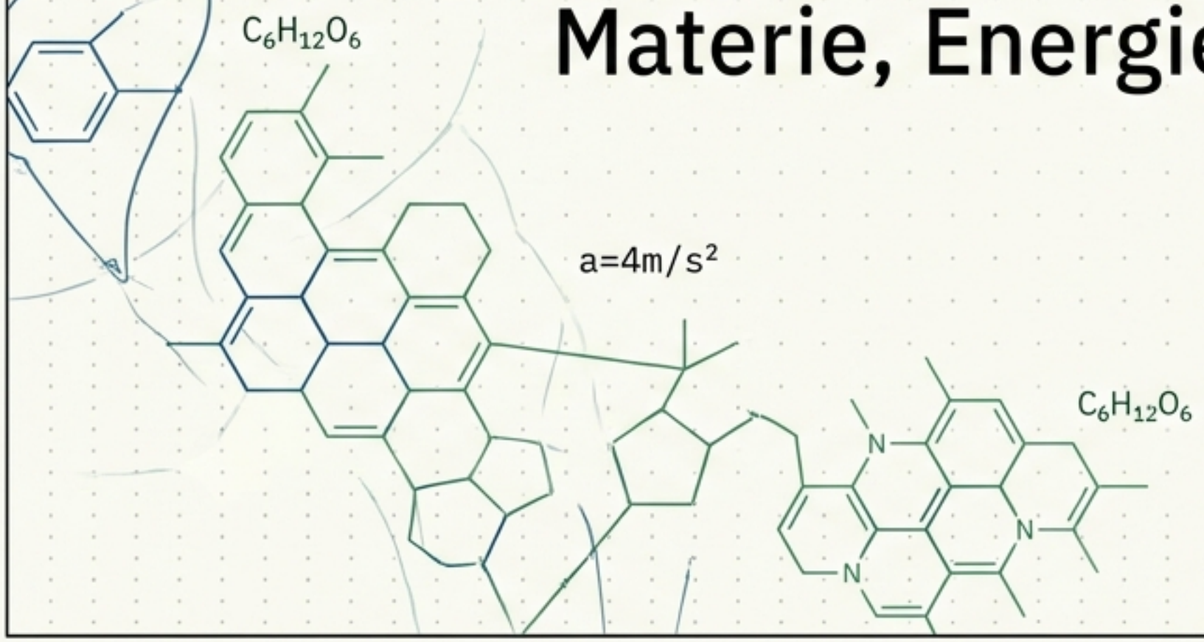
Exam Date: 09.03.2026

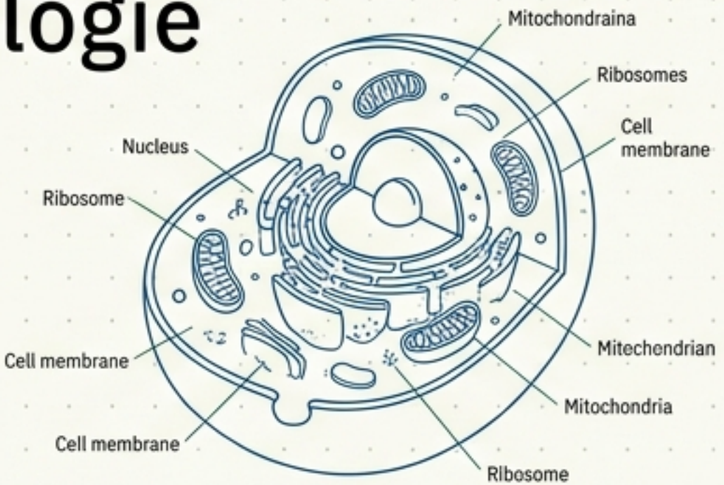




NW LZK Study Deck: Class 1IBK

Materie, Energie, Kräfte, Biologie





Made by VSN (VBS Student Network)

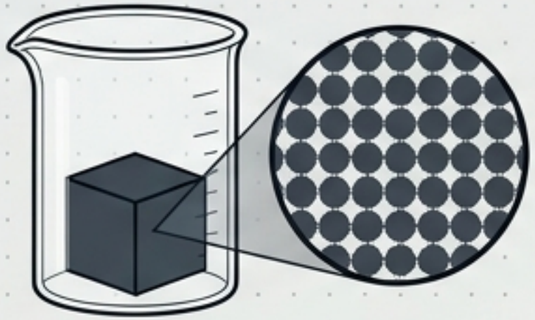
Gemini Notebook

Module 1: Matter

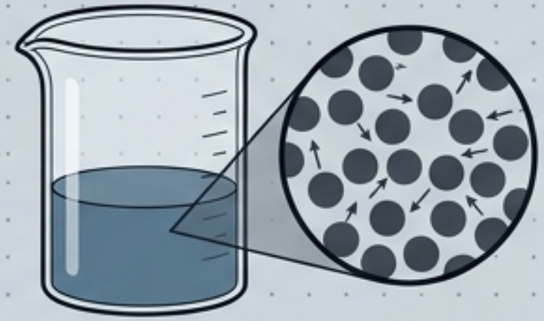
Adding or removing heat drives the change. Plasma is unique—look out for **“ionized”** or **“charged particles”** as the keyword!

State of matter *(Aggregatzustand)*

Solid *(fest)*
Shape: Constant | Volume: Constant

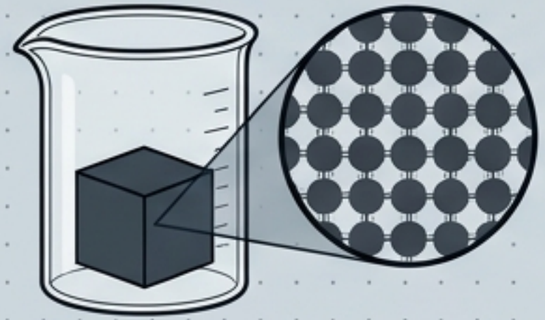


Liquid *(flüssig)*
Shape: Adapts | Volume: Constant

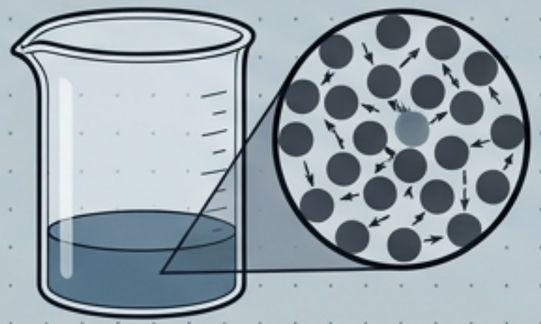




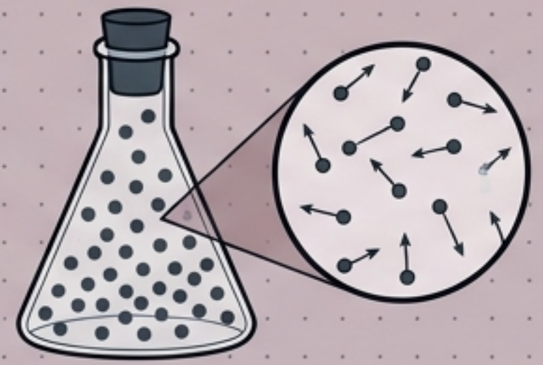
Solid *(fest)*
Shape: Constant | Volume: Constant



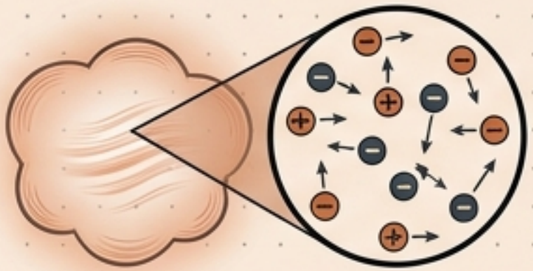
Liquid *(flüssig)*
Shape: Adapts | Volume: Constant



Gas *(gasförmig)*
Shape: Adapts | Volume: Variable

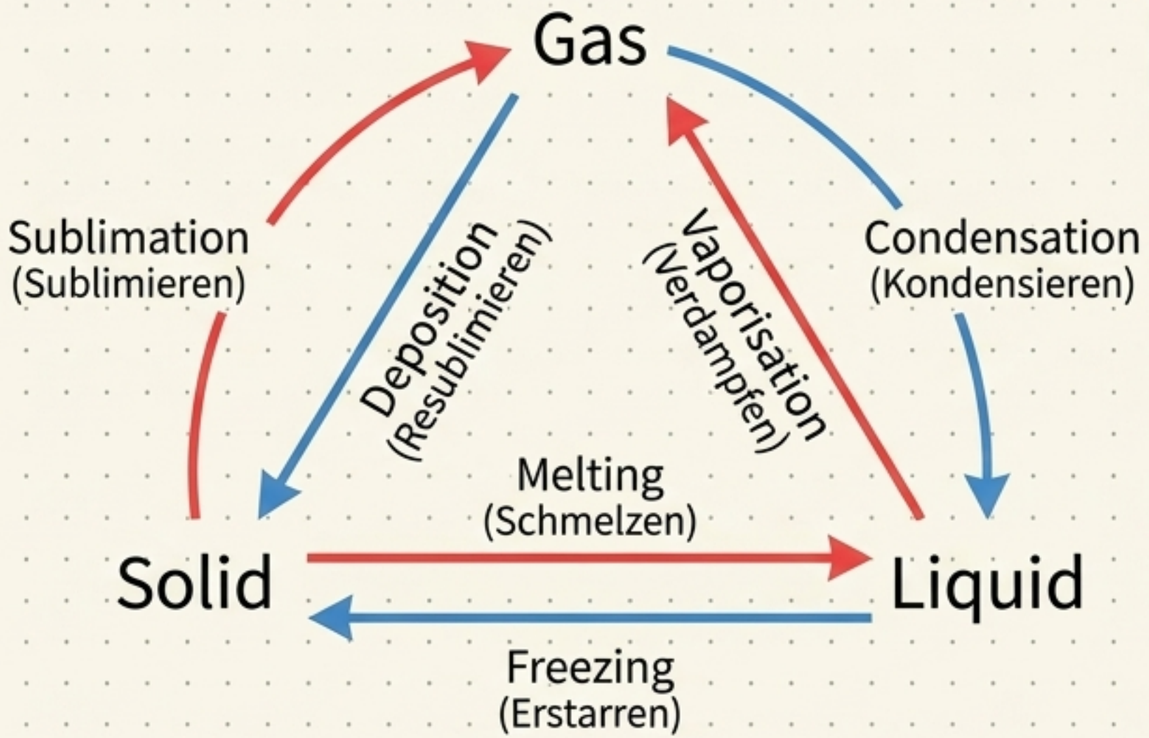


Plasma *(plasmaförmig)*
Shape: Variable | Volume: Variable



Gemini Notebook

Sublimation (Solid → Gas) skips the liquid phase entirely. Memorize this exception!

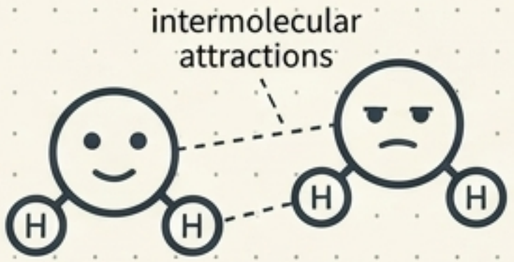


Scratchpad

Ü 3.7 The Particle Dance



Gas



Condensing

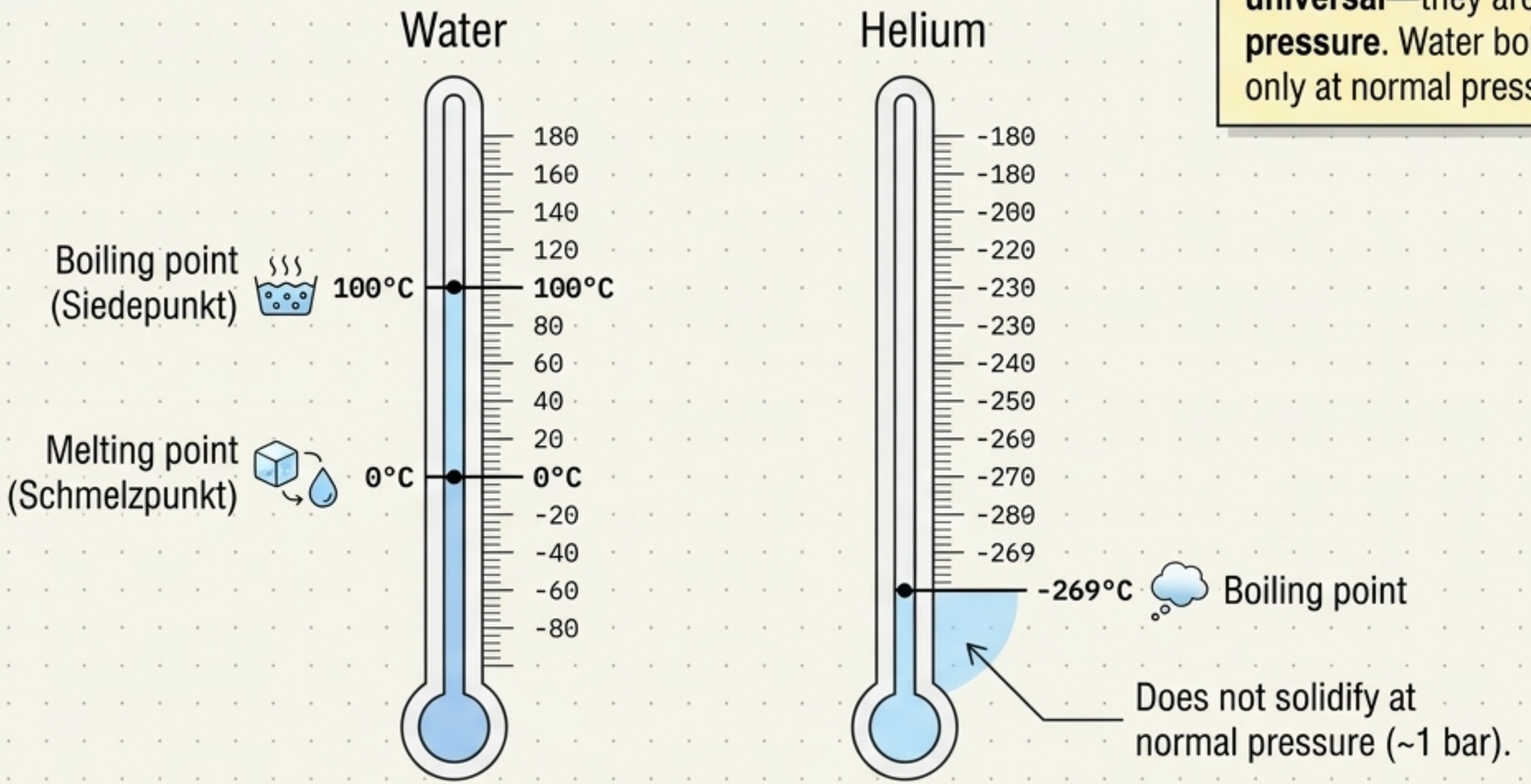


Freezing

Gemini Notebook

Module 1: Matter

Phase temperatures are **NOT universal**—they are tied to **pressure**. Water boils at 100°C only at normal pressure (1013 hPa).



Scratchpad

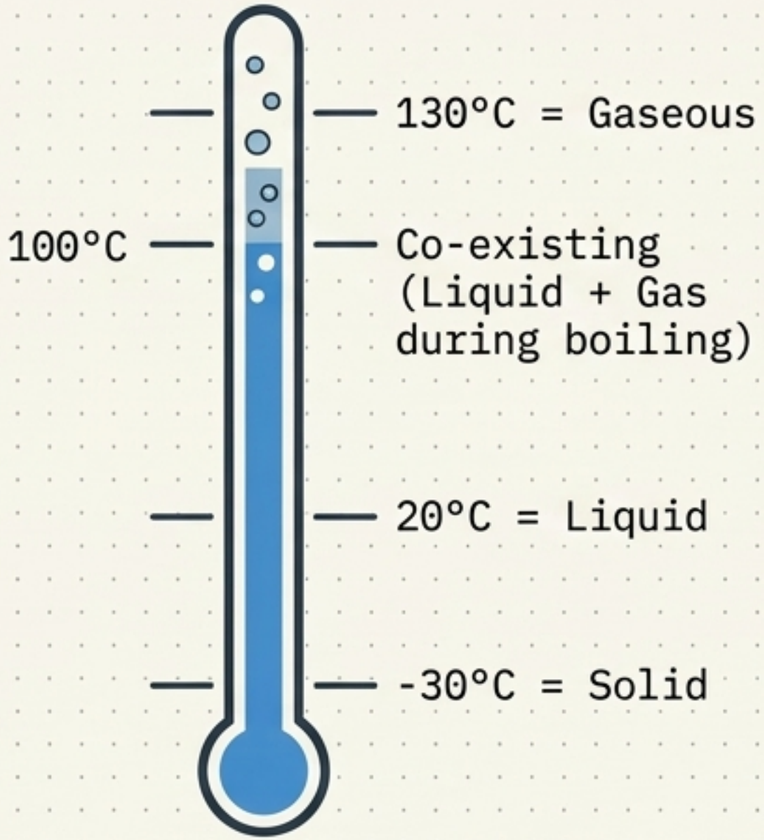
```
Ü 3.19 Hydrogen (Wasserstoff):
Melting: -259°C | Boiling: -253°C
Q: State at 20°C? -> Gaseous.
Q: When is it liquid? -> Only between -259°C and -253°C.
```

Gemini Notebook

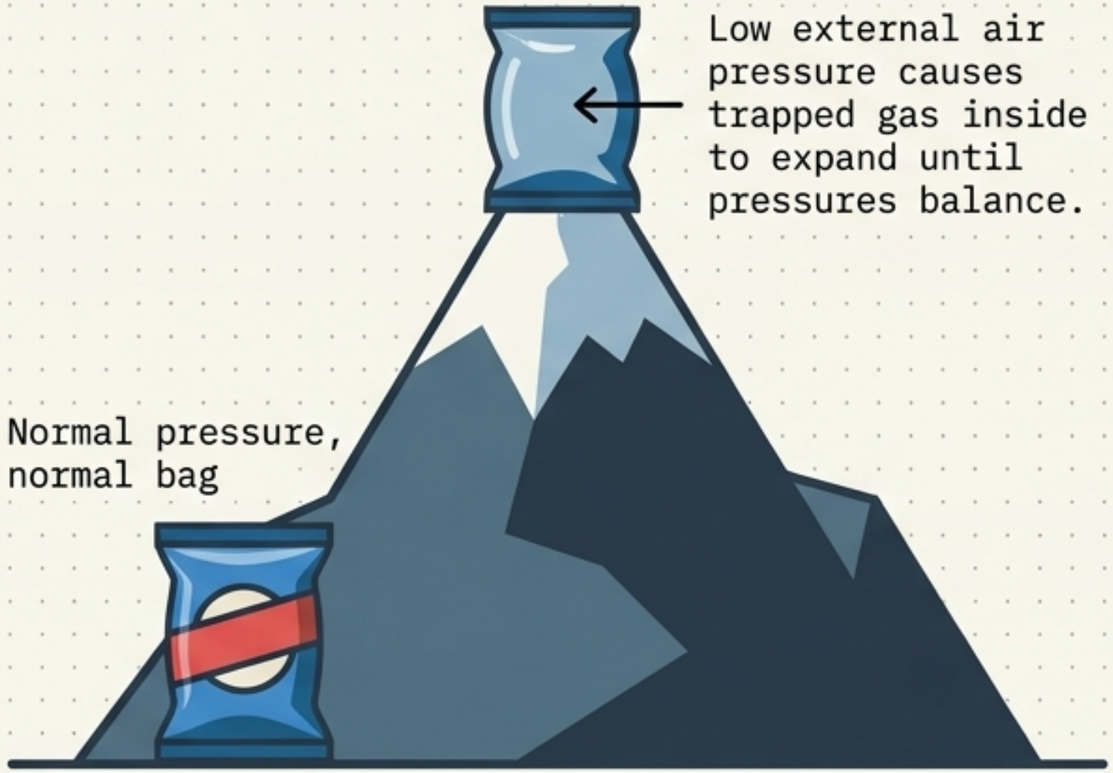
Module 1: Matter

If asked about the chip packet, do NOT write that the chips produce gas. It is strictly a pressure/volume effect!

Ü 3.18 Water States



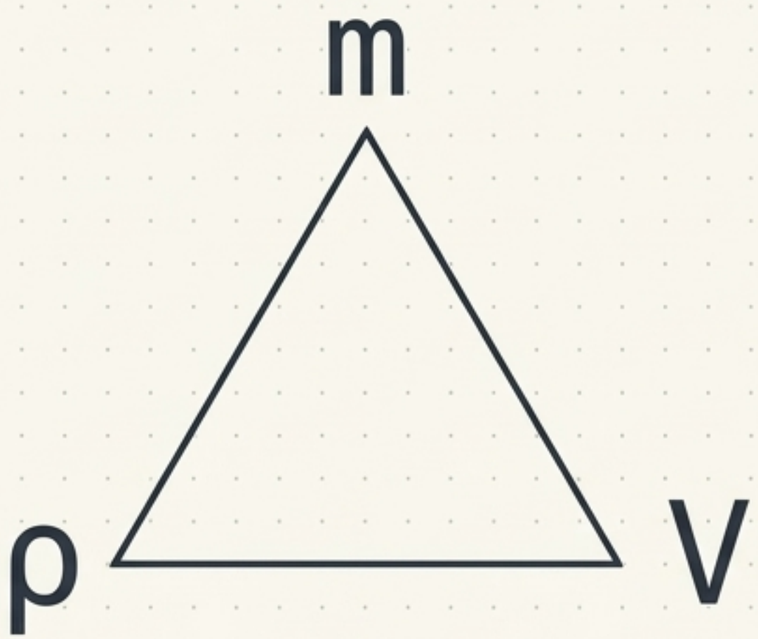
Ü 3.20 Chip Packet



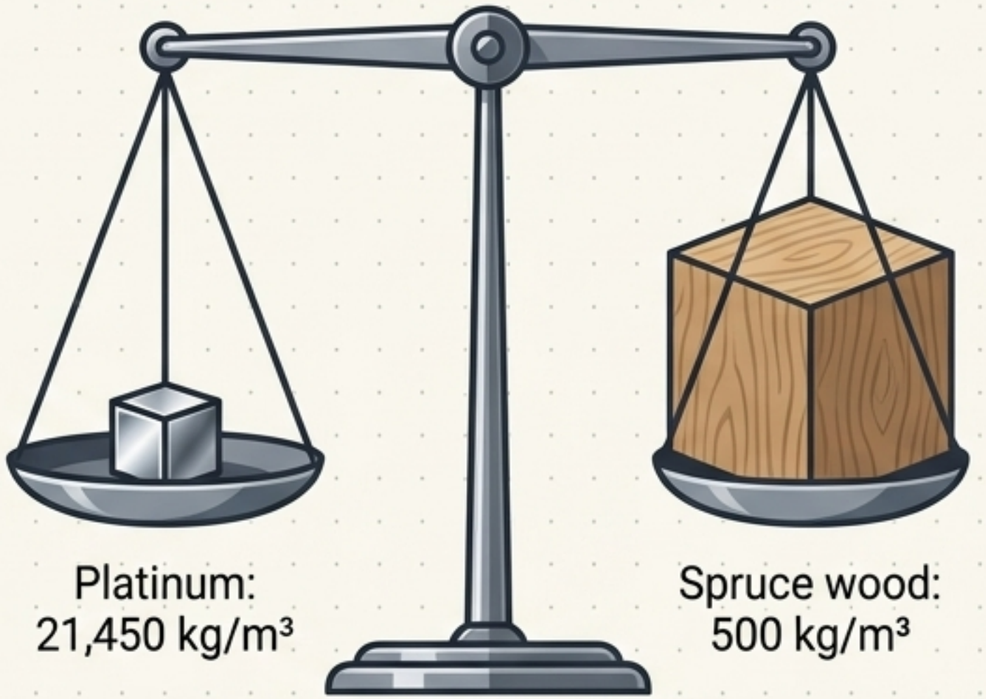
Gemini Notebook

Module 2: Density

Before calculating, ALWAYS check your units.
1 g/cm³ is exactly equal to 1000 kg/m³.



Mass density (Massendichte)	ρ
Mass (Masse)	m
Volume (Volumen)	V



Equal Mass, Different Volume

Scratchpad

Ü 3.8 Regular Bodies: Volume of a rectangular solid = length × width × height ($V = l \times w \times h$).
Measure mass with a scale, then apply $\rho = m/V$.

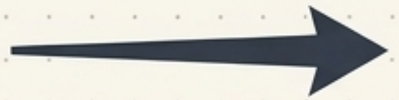
Gemini Notebook

Module 2: Density

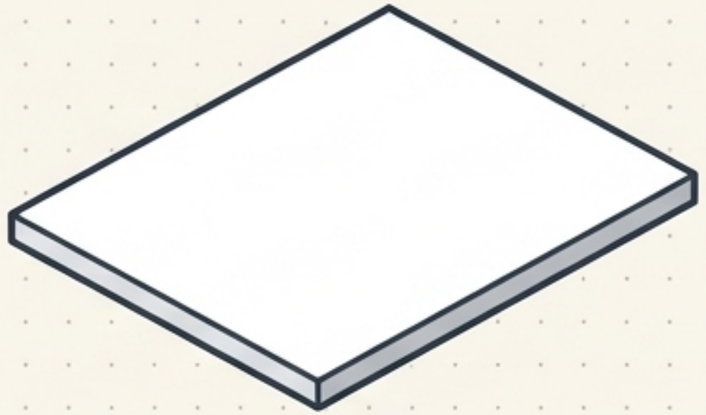
Most things get denser when cold because particles move closer together. Water is the exception—ice floats because it is less dense than liquid water!



Steel: 1 dm^3, 7.85 kg



Equal mass, unequal volume



Polystyrene: 0.3925 m^3, 7.85 kg

Scratchpad

Ü 3.9 Room-air Mass:

Room V = 9 × 7 × 3.6 = 226.8 m^3. Air density ρ = 1.2 kg/m^3. Mass m = ρ × V = 1.2 × 226.8 = 272.16 kg.

Ü 3.10 Steel vs Polystyrene:

Steel mass = 7.85 kg. Polystyrene ρ = 20 kg/m^3.
Required volume for same mass: V = 7.85 / 20 = 0.3925 m^3.
With a base of 2 m^2, height = 0.3925 / 2 = 0.196 m (19.6 cm).

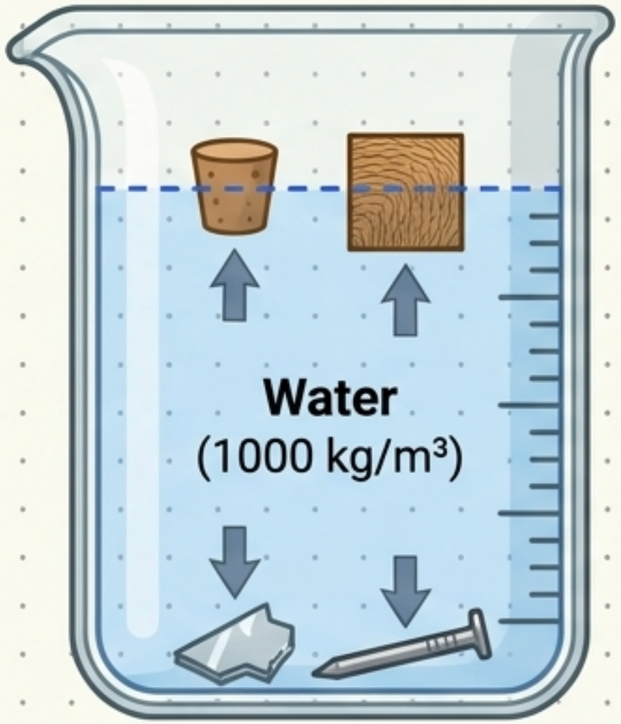
Gemini Notebook

Module 2: Density

Materials with density < 1000 kg/m³ float in water. 1000 is the magic number to memorize.

Density Ranking (kg/m³)

Oil (200 kg/m³)
Cork (204 kg/m³)
Wood (700 kg/m³)
Polyethylene (860 kg/m³)
WATER (1000 kg/m³)
Glass (2200 kg/m³)
Stone (4000 kg/m³)
Iron (89,000 kg/m³)



Scratchpad

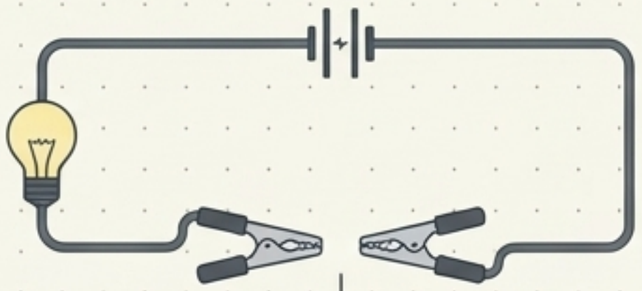
Ü 3.39 Salt Solution: 100 ml = 100 cm³. ρ = 109g / 100cm³ = 1.09 g/cm³ (which equals 1090 kg/m³).

Gemini Notebook

Module 2: Conductivity

Metals conduct via mobile electrons; electrolytes via ions. Semiconductors (like Silicon) conduct better when heated!

Conductivity (Leitfähigkeit)



Outcome 1



Bright lamp = **Good Conductor** (Leiter).
Example: Silver, Copper, Aluminum.

Outcome 2



Dim lamp = **Semiconductor** (Halbleiter) / Moderate conductor.

Outcome 3



Dark lamp = **Insulator** (Isolator/Nichtleiter).
Example: Rubber, Glass, Distilled Water.

Scratchpad

Gemini Notebook

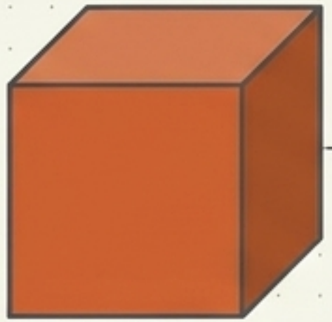
Module 3: Energy

Energy cannot be created or destroyed, only transformed. This is the **Law of Conservation of Energy**.

Energy (Energie)

Work (Arbeit)

1 J = 1 Nm = 1 Ws



The energy required to heat just 1 kg of water by 1°C

1 J

4190 J

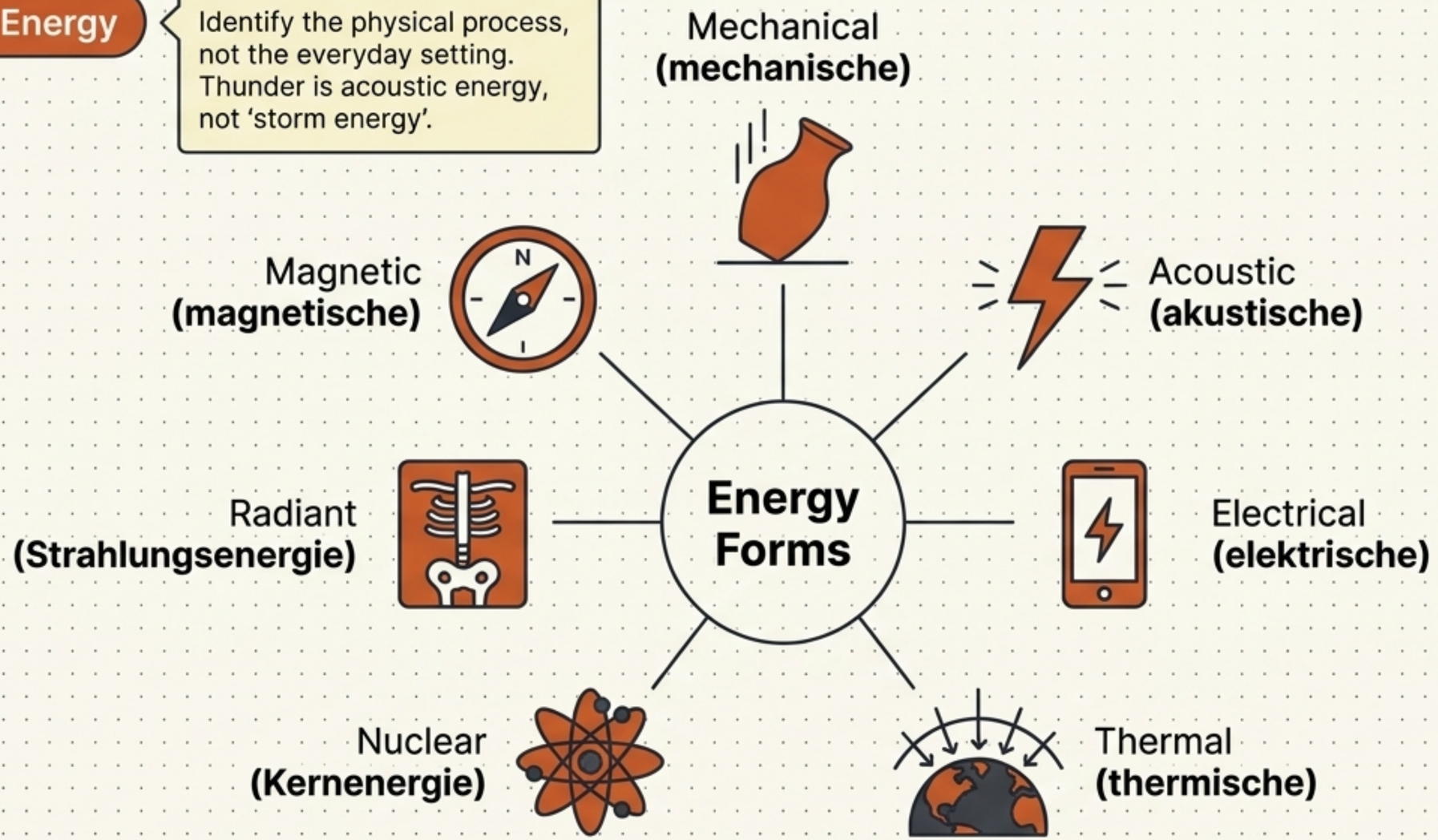
Scratchpad

Definition: Energy is the ability to do work. Named after James Prescott Joule.

Gemini Notebook

Module 3: Energy

Identify the physical process, not the everyday setting. Thunder is acoustic energy, not ‘storm energy’.



Scratchpad

Ü 3.17 Hardening: Wet clay -> irreversible kiln drying (water escapes).
Iron/steel -> heating + rapid quenching.

Gemini Notebook

Module 4: Forces

Force (Kraft)
Acceleration (Beschleunigung)
Deformation (Verformung)

Force is a vector quantity—it MUST have **a direction**. Always draw an arrow to represent it, with length showing magnitude.

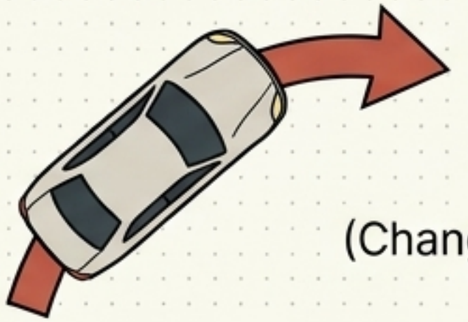
Speeding up
(Car accelerating)



Slowing down
(Negative acceleration / braking)



$$\vec{F} = m\vec{a}$$



Turning
(Changing direction)

Deforming
(Crushing a can)



Scratchpad

$1\ \mathrm{N} = 1\ \mathrm{kg} \cdot \mathrm{m/s^2}$.

Gemini Notebook

Module 4: Forces

In Axiom 3, the forces are equal and opposite, but they act on different bodies, so they don't cancel each other out on a single object.

Newton's Axioms (Newtonsche Axiome)

1. Inertia (Trägheit)

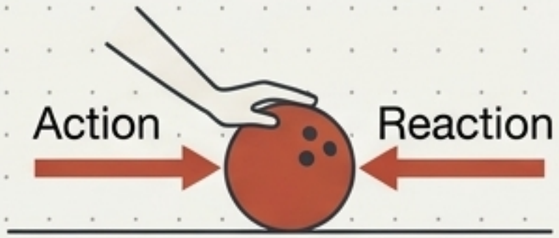


Body wants to keep its previous state of motion.

2. F = ma



3. Action-Reaction (Wechselwirkung)



Scratchpad

$F = ma$

$\sum F = 0 \rightarrow v = \text{constant}$

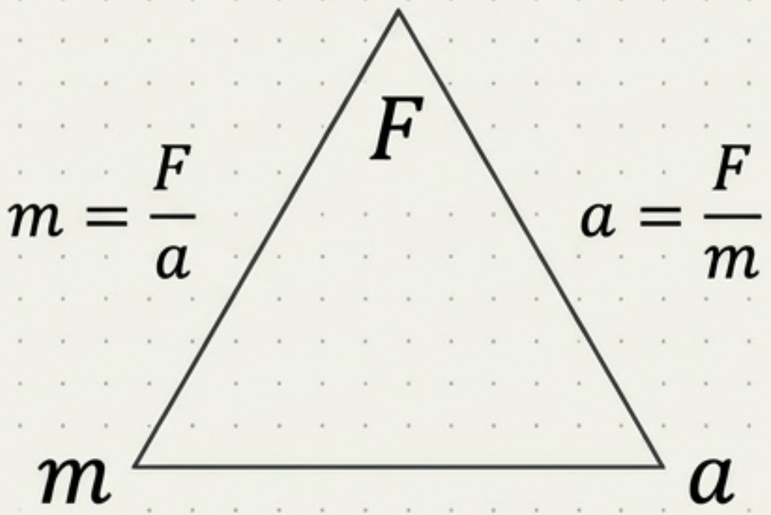
$F_{action} = -F_{reaction}$

Gemini Notebook

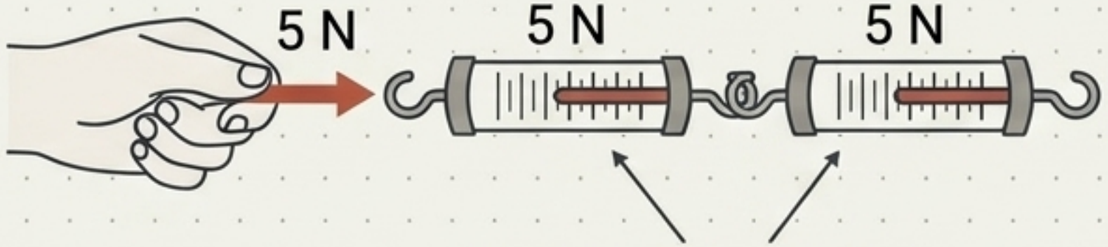
Module 4: Forces

Double the mass needs double the force for the exact same acceleration (Ü 3.28).

Algebraic manipulation triangle



V 3.5 Series Springs



Tension is transmitted, not split.

Scratchpad

Ü 3.25: F = 3 × 2 = 6 N. a = 6 / 12 = 0.5 m/s².

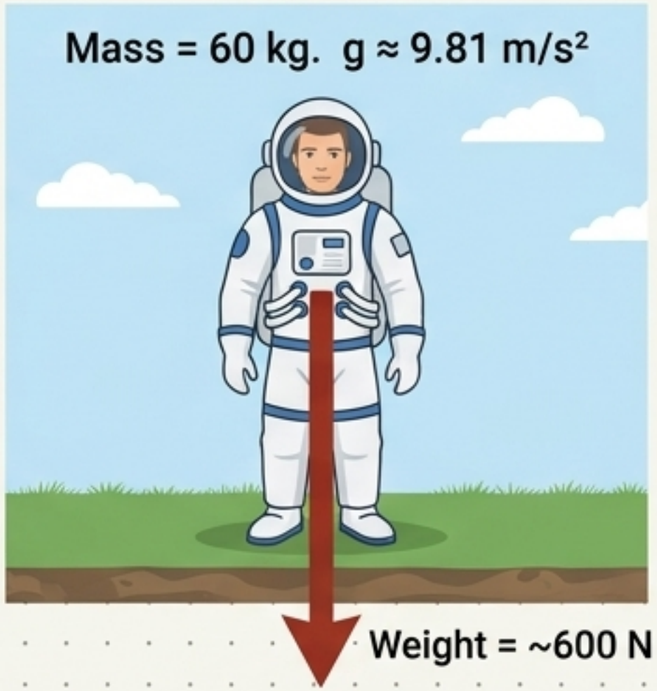
Ü 3.26 (m=10kg): If F=5N, a = 0.5 m/s². If F=10N, a = 1 m/s².

Gemini Notebook

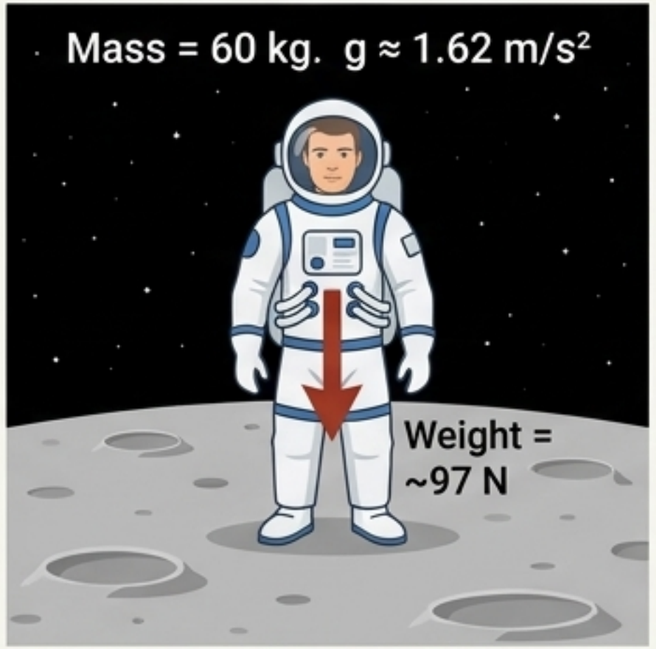
Mass (kg) is an intrinsic property and NEVER changes. Weight force (N) depends entirely on where you are in the universe.

Gravitational force / Weight force (Gravitationskraft / Gewichtskraft)

On Earth



On Moon (Ü 3.29)



Scratchpad

Ü 3.40 Conversions (using g=10):
1 kg -> 10 N.
10 dag (0.1 kg) -> 1 N.
1 g -> 0.01 N.

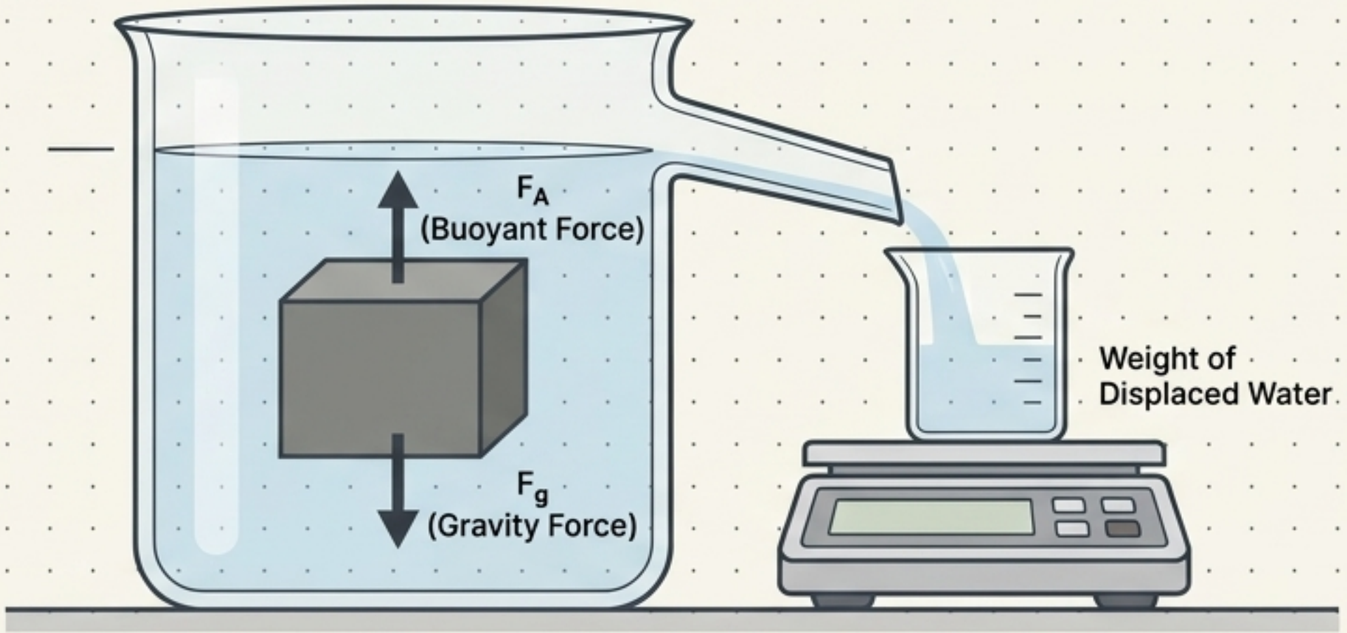
Gemini Notebook

Buoyancy equals the weight of the fluid pushed out of the way. It depends on the fluid's density, not the object's mass!

Buoyant force (Auftriebskraft)

Displaced (verdrängt)

$$F_A = \rho_{medium} \times V_{displaced} \times g$$



Scratchpad

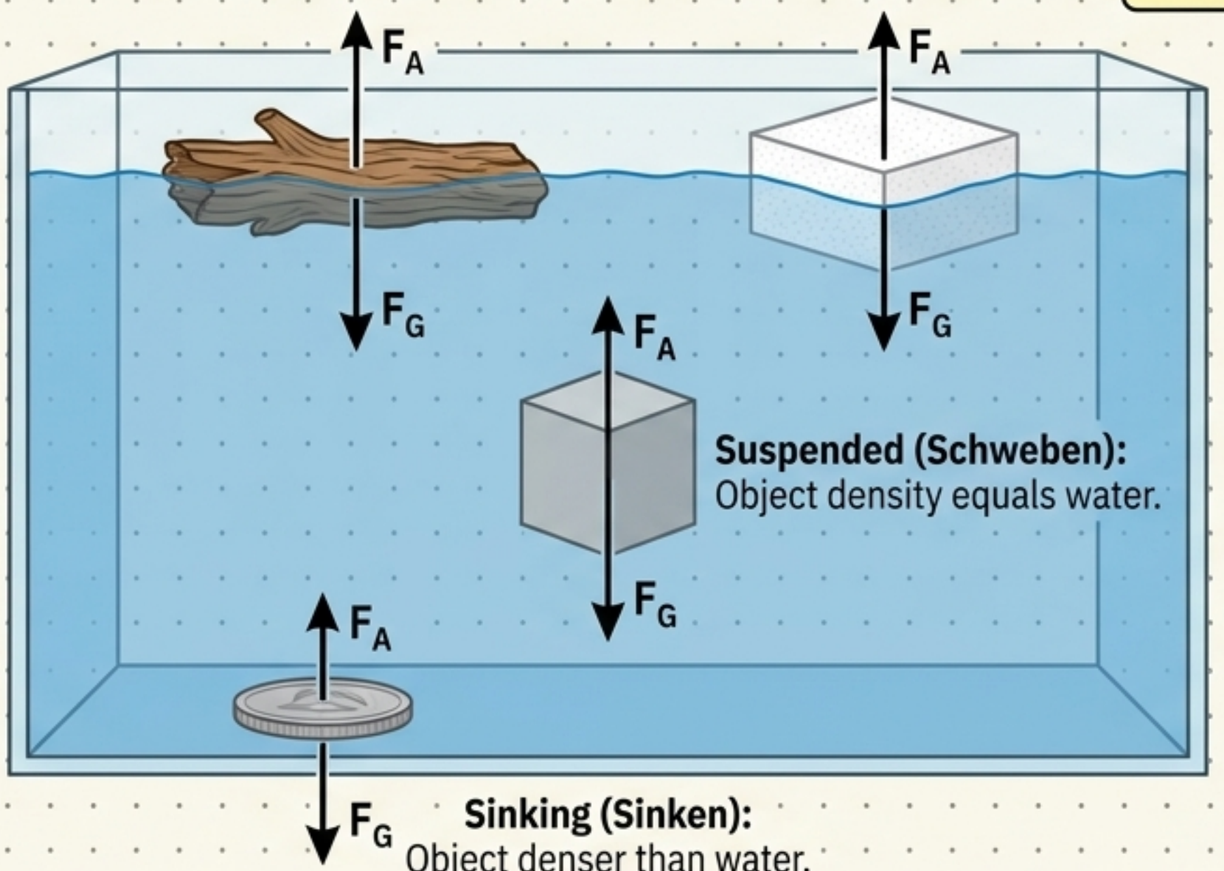
Rechenbeispiel 13.1: Volume = 5 cm^3 = 0.000005 m^3. F_A = 0.000005 × 1000 (water) × 10 = 0.05 N.

Gemini Notebook

Module 5: Gravity & Buoyancy

Floating (Schwimmen/Steigen):
$F_A = F_G$ at surface equilibrium (partial displacement).

If an object is floating at rest at the surface, F_A exactly equals F_G.



Suspended (Schweben):
Object density equals water.

Sinking (Sinken):
Object denser than water.

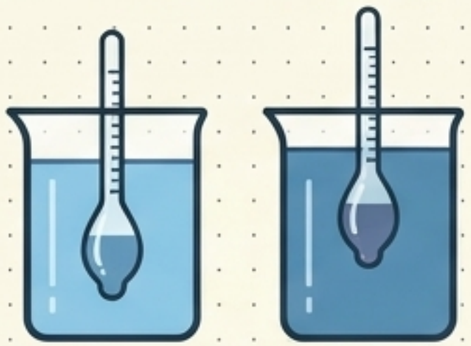
Scratchpad

K 3.18 Driftwood: Cube 0.1m × 0.1m × 0.1m. Mass 0.7 kg, F_G = 7 N. Submerged volume displaces exactly 7 N of water (0.0007 m^3).

Gemini Notebook

Module 5: Gravity & Buoyancy

Convert volume to m^3 before calculating! 1000 dm^3 = 1 m^3.

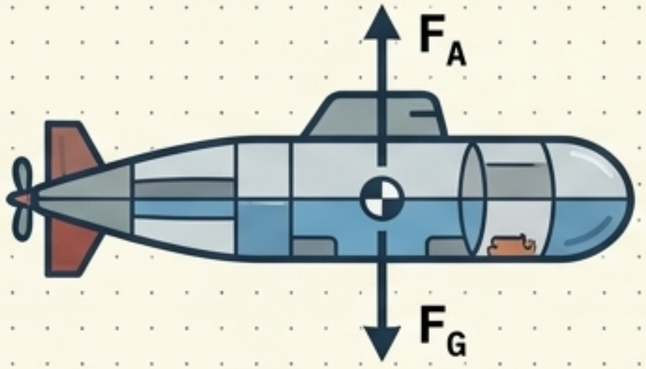


1. Hydrometer (K 3.12)

Floats deeper in lower-density liquids.

In water (1000 kg/m^3), 90% submerged.

In denser sugar water (1050 kg/m^3), only ~85.7% submerged.

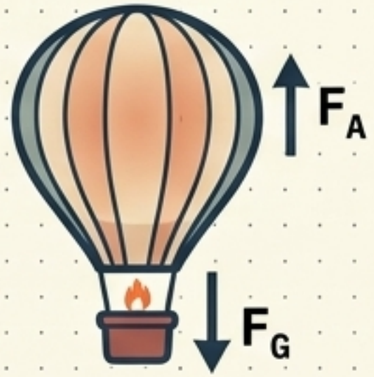


2. Submarine (Ü 3.32)

Changes density via ballast tanks.

If F_G = 5000 N, hovering requires F_A = 5000 N.

Volume V = 5000 / (1000 × 10) = 0.5 m^3.



3. Hot Air Balloon (K 3.19)

Internal hot air is less dense than outside air.

Net lift = 1 N/m^3. To lift 2000 N, need 2000 m^3 volume.

Gemini Notebook

1. Zoology	2. Botany
Animals	Plants
3. Microbiology	**4. Ecology**
Microorganisms	Relationships

Always capitalize the Genus and lowercase the species. If handwriting on the exam, underline the name to indicate italics!

Binomial nomenclature (binäre Nomenklatur)

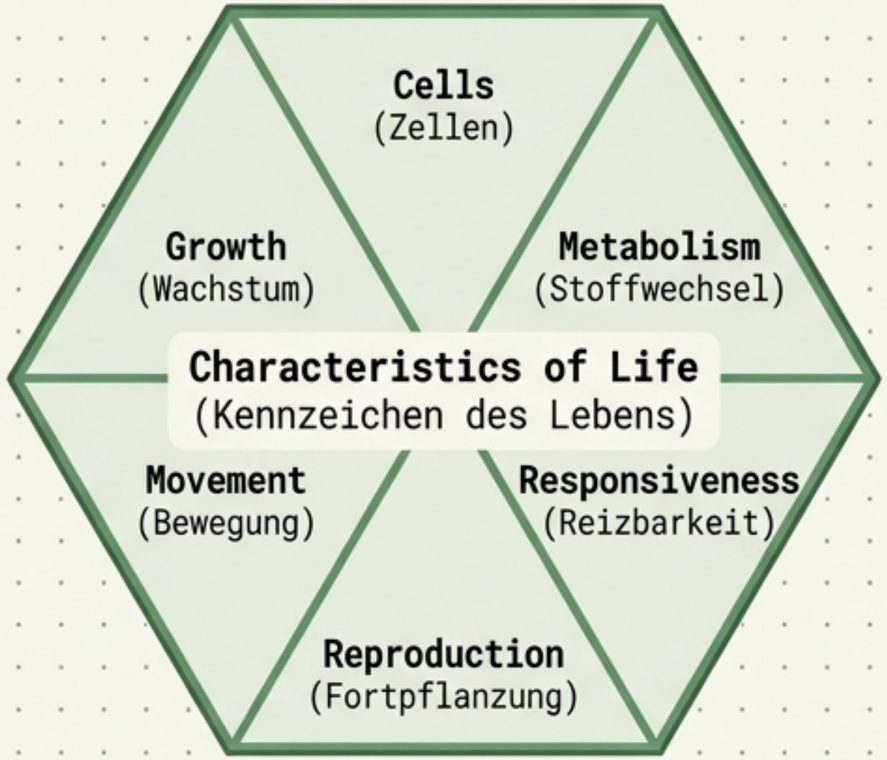
Homo sapiens

Genus (Gattung)

Species epithet (Art-Epitheton)

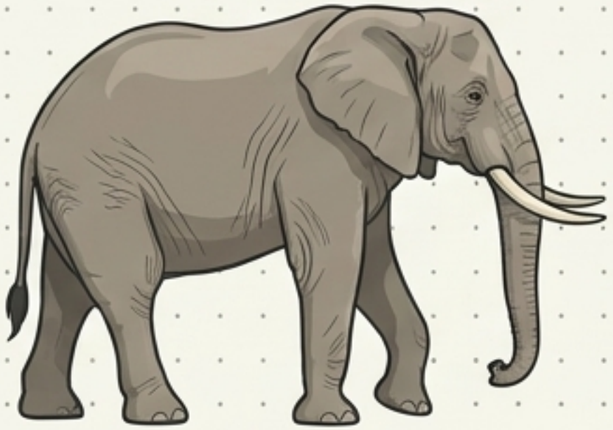
Gemini Notebook

Module 6: Biology



Movement alone does not prove life (e.g., a car). An organism MUST have the full set of characteristics.

Ü 4.1 Real vs. Toy Elephant



- ✔ cellular
- ✔ metabolizes
- ✔ grows
- ✔ responds
- ✔ reproduces



- ✘ cellular
- ✘ metabolizes
- ✘ grows
- ✘ responds
- ✘ reproduces

Gemini Notebook

Exam Blueprint: Master Cheat Sheet

Write these formulas down at the top of your test paper the second the exam begins! You are ready for 09.03.2026. Viel Erfolg!

Density:	$\rho = m/V \mid 1\ g/cm^3 = 1000\ kg/m^3$
Volume:	$1\ m^3 = 1000\ dm^3 = 1000\ L$
Energy:	$1\ J = 1\ Nm = 1\ Ws$
Force:	$F = m \cdot a \mid 1\ N = 1\ kg \cdot m/s^2$
Gravity:	$F_G = m \cdot g$ (Earth $g \approx 10\ m/s^2$)
Buoyancy:	$F_A = \rho_{med} \cdot V_{disp} \cdot g$
Float/Sink:	Float < 1000 kg/m³ < Sink

Gemini Notebook